
Fairness Auditing of an Income-Classification Model: SHAP-Based Analysis of Proxy Features

Author One Affiliation One author.one@example.com	Author Two Affiliation Two author.two@example.com	Author Three Affiliation Three author.three@example.com
Author Four Affiliation Four author.four@example.com	Author Five Affiliation Five author.five@example.com	

Abstract

to be determined

1 Introduction

Machine-learning models can produce unequal outcomes across demographic groups even when their average predictive performance is high [Mehrabi et al., 2021]. A common response is to remove a sensitive attribute, such as `sex`, from the model inputs. This intervention is simple and often presented as a way to prevent discrimination. However, other features may still contain information associated with the protected attribute and allow the model to reproduce group disparities [Datta et al., 2017]. Removing `sex` therefore does not by itself establish that a model is fair.

This paper studies that problem in income classification. We use the UCI Adult Income dataset, which contains demographic, educational, occupational, and work-related attributes and a binary target indicating whether annual income exceeds \$50,000 [Dua and Graff, 2019]. The dataset provides a useful setting for a controlled audit because it contains an explicit `sex` attribute alongside several features that may encode social and economic structure. Nevertheless, the dataset reflects a particular historical collection process, uses a binary representation of `sex`, and cannot support universal conclusions about fairness in income prediction. Our conclusions are therefore limited to the observed groups, features, model choices, and audit design.

We ask whether excluding `sex` changes the fairness of an income-classification model and whether remaining features can act as proxies for it. To answer this question, we compare models trained with and without `sex`, investigate whether selected remaining features contribute to group-related disparities, and compare feature removal with fairness-aware mitigation. Predictive performance and group fairness are evaluated together so that an apparent fairness improvement is not interpreted without considering its effect on utility.

An overview of our method is provided in Figure 1. This paper makes three contributions:

- We provide a reproducible comparison of income-classification models trained with and without the explicit sensitive attribute `sex`.
- We combine model explanations, subgroup comparisons, and feature ablation to investigate whether specific remaining features behave as potential proxies.
- We compare proxy removal with fairness-constrained learning using common predictive and group-fairness metrics, making the associated performance–fairness trade-offs explicit.

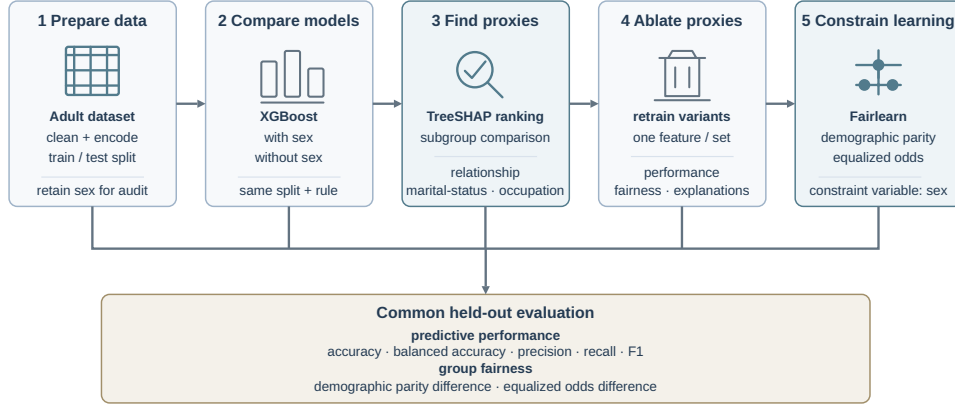


Figure 1: Overview of our proposed method. The audit compares explicit-feature removal, proxy ablation, and fairness-constrained learning under a common held-out evaluation.

The central contribution is not a claim that any one feature is inherently unfair or that one fairness metric is sufficient. Rather, we provide an audit of whether removing an explicit sensitive attribute is enough to change observed disparities when related information remains available to the model.

The remainder of this paper is structured as follows. Section 2 discusses related work, ...

2 Related Work

Fairness through unawareness, proxy discrimination, and causal fairness. One influential response to discriminatory prediction is to omit protected attributes from the model. However, Dwork et al. [2012] argue that fairness cannot be reduced to ignoring sensitive attributes: their *Fairness Through Awareness* framework instead requires similar individuals, under a task-specific similarity metric, to receive similar treatment. This distinction is directly relevant to our comparison between models with and without `sex`: removing the column is an unawareness baseline, not evidence that the remaining representation is fair. The proxy discrimination literature makes the same concern operational by studying how protected-group information can be inferred from seemingly permissible features and how such relationships can be audited [Datta et al., 2017]. By contrast, Kusner et al. [2017] define counterfactual fairness through a causal model, requiring a prediction to remain unchanged in counterfactual worlds in which an individual’s protected attribute differs. Our SHAP rankings, subgroup comparisons, and ablations are deliberately more modest: they can identify predictive association and model dependence on candidate proxies, but they cannot determine whether a feature is a causal mediator, a legitimate correlate, or an impermissible proxy.

Group fairness criteria and the problem of incompatible objectives. A second body of work formalizes how disparities should be measured once a model has been trained. Hardt et al. [2016] introduce equality of opportunity and equalized odds, connecting fairness to conditional error rates and providing a post-processing procedure that uses group membership to alter predictions. These ideas motivate our reporting of true- and false-positive rate disparities in addition to selection-rate disparities. Yet no single criterion resolves the broader normative question. Chouldechova [2017] show that calibration and error-rate parity can conflict when outcome prevalence differs between groups, while Kleinberg et al. [2017] prove closely related impossibility results for risk scores. Critical reviews likewise emphasize that criteria such as statistical parity encode different values and can lead to different conclusions about the same system [Corbett-Davies and Goel, 2017, Barocas et al., 2019]. Accordingly, our audit reports demographic-parity difference and equalized-odds difference as separate descriptive outcomes alongside accuracy, balanced accuracy, precision, recall, and F1. The Adult experiment therefore tests whether an intervention changes particular measured disparities; it does not treat a small value for one metric as a complete definition of fairness.

Explainability and feature-level auditing for proxy detection. The third relevant line of work uses explanations to inspect how models reach their predictions. Lundberg and Lee [2017] introduce SHAP, an additive feature-attribution framework grounded in Shapley values whose local accuracy and consistency properties make it attractive for comparing contributions across individual predictions. We use TreeSHAP in this spirit to rank features in the sex-excluded XGBoost model and to prioritize variables such as `relationship`, `marital-status`, `occupation`, and `hours-per-week` for further investigation. SHAP, however, is an explanatory device rather than a fairness test: a large attribution indicates model reliance, not discriminatory intent or causal influence. This caveat is especially important for Adult because its predictors are dependent. Aas et al. [2021] show that common SHAP approximations can be misleading when feature independence assumptions fail and propose dependence-aware approximations and grouped explanations. Our use of explanations is therefore paired with subgroup comparisons and feature ablation. The combination asks a narrower and testable question than explanation alone: whether removing a candidate feature changes both model behavior and observed group disparities on the same held-out population.

Fairness-aware mitigation and constrained learning. Feature deletion is only one possible intervention, and it may remove useful predictive information without directly controlling the disparity of interest. Zafar et al. [2017] incorporate fairness constraints into the geometry of a classifier’s decision boundary, making the fairness–accuracy trade-off an explicit part of training. Madras et al. [2018] instead learn representations that make sensitive attributes difficult to predict using an adversary, relating the training objective to demographic parity, equal opportunity, and equalized odds. The reductions approach of Agarwal et al. [2018] is particularly close to our implementation: it converts constrained binary classification into a sequence of cost-sensitive classification problems and can optimize error subject to demographic-parity or equalized-odds constraints. This provides the conceptual basis for Fairlearn’s `ExponentiatedGradient` comparison in our audit. Our design places these approaches beside, rather than above, sex removal and proxy ablation: all variants use the same fixed split and held-out metrics, allowing us to distinguish fairness obtained by deleting features from fairness obtained by explicitly constraining predictions. The comparison also preserves sex for evaluation and mitigation constraints while excluding it from ordinary predictive features, so that the fairness intervention is measurable without treating the sensitive attribute as a standard model input.

Taken together, prior work establishes four points that motivate our study: omitting a sensitive feature does not rule out proxy discrimination; fairness criteria measure different and sometimes incompatible properties; explanations can reveal model reliance but do not by themselves establish causality or unfairness; and constrained learners offer a direct alternative to feature deletion. The gap addressed here is an integrated, reproducible audit that connects these pieces in one Adult-income experiment: it compares explicit sex removal with targeted proxy ablation, uses SHAP and subgroup diagnostics to justify which proxies are tested, and evaluates both feature interventions and Fairlearn reductions under common predictive and group-fairness metrics. This makes the practical claim more precise than either “remove the sensitive attribute” or “optimize a fairness metric”: the question is which intervention changes which observed disparities, at what performance and explainability cost, under the stated dataset and audit limitations.

3 Methodology

3.1 Our proposed method

We propose a reproducible, diagnostic audit protocol for testing whether removing an explicit sensitive attribute is sufficient to reduce group disparities. The protocol treats feature deletion as one intervention in a larger comparison rather than as evidence of fairness by itself. It has five stages: (i) construct a leakage-controlled preprocessing and evaluation pipeline, (ii) compare models with and without sex, (iii) identify candidate proxy features using explanations and subgroup diagnostics, (iv) measure the effect of ablating those candidates, and (v) compare feature-based interventions with constrained learning. The sensitive attribute is retained outside the model feature matrix throughout the experiment so that it can be used for subgroup evaluation.

Leakage-controlled pipeline. The supplied Adult training and test files are cleaned with one deterministic ETL procedure. Categorical variables are represented explicitly, including the documented

missing-value category, and the income label is mapped to a binary target. Preprocessing components such as category encoding and numeric transformations are fitted on the training split only and then applied to the held-out test split. The target and `sex` are never used to fit preprocessing for the model features. This separation prevents information from the evaluation split from influencing training while preserving the attribute needed for fairness measurement.

Model comparison. The primary predictive model is an XGBoost classifier trained under a fixed configuration and evaluated with the same decision rule for every variant. We first fit a baseline using the full permitted feature set, including `sex`. We then fit a sex-excluded model after removing only `sex` from the feature matrix. This paired comparison isolates the effect of explicit attribute removal while holding the data split, preprocessing, model family, and evaluation procedure constant. Additional variants remove suspected proxies in a prespecified sequence, including `relationship`, `marital-status`, `occupation`, and `hours-per-week`. We report every attempted variant and do not interpret a feature’s removal as proof that the feature is intrinsically unfair.

Experimental setup. The baseline uses the official Adult training file (32,561 records) for fitting and the official Adult test file (16,281 records) for final evaluation; no additional random split is drawn. The model input contains the six numeric predictors and eight categorical predictors listed in the dataset, including `sex`. Numeric columns are passed through and categorical columns are one-hot encoded with unknown test categories ignored. The transformer is fitted on the training split only. The resulting baseline representation contains 108 transformed features. We use XGBoost with 300 trees, maximum depth 6, learning rate 0.1, histogram tree construction, and random seed 42; subsampling and column subsampling are both 1.0, and the training objective is binary logistic classification with log-loss evaluation. Predictions use the positive-class probability threshold 0.5. The saved test predictions retain `sex` and the observed income label solely for post-hoc evaluation.

Proxy diagnostics. Proxy analysis is performed only after fitting the sex-excluded model. We use TreeSHAP to rank feature contributions to individual predictions [Lundberg and Lee, 2017], summarize the rankings over the evaluation population, and compare feature distributions and prediction rates across the observed `sex` groups. Features are carried forward as candidate proxies when their model contribution and subgroup association provide convergent evidence in this dataset. We then perform feature ablation by retraining the model without one candidate or a prespecified candidate set. The ablation comparison distinguishes three quantities: predictive change, change in group metrics, and change in the explanatory profile. Because these diagnostics are observational and model-dependent, they identify candidate proxy pathways rather than establish causality.

Fairness-constrained comparison. As a complementary intervention, we train Fairlearn ExponentiatedGradient models with the broader feature representation and constraints based on demographic parity or equalized odds [Agarwal et al., 2018]. The sensitive attribute is supplied to the mitigation procedure as a constraint variable, not as an ordinary predictive feature. This comparison tests whether an explicit group constraint changes the performance–fairness trade-off differently from deleting features. The unconstrained baseline, sex-excluded model, proxy-ablation variants, and constrained models are evaluated on the same held-out data and summarized in one comparison table.

Decision and reporting rule. For each variant, we report accuracy, balanced accuracy, precision, recall, F1, demographic-parity difference, and equalized-odds difference. Lower fairness-difference values indicate smaller observed disparities under the corresponding definition, but no single metric is treated as a sufficient fairness criterion. The protocol therefore reports predictive and fairness results separately, records feature sets and preprocessing decisions, and interprets changes only within the Adult dataset, selected model family, binary `sex` representation, and fixed prediction rule.

3.2 Dataset

We use the UCI Adult Income dataset [Dua and Graff, 2019], comprising 48,842 records and 14 predictors. The binary target is whether annual income exceeds \$50,000, and `sex` is retained as a sensitive attribute for evaluation. We preserve the dataset’s supplied training and test files rather than drawing a new split.

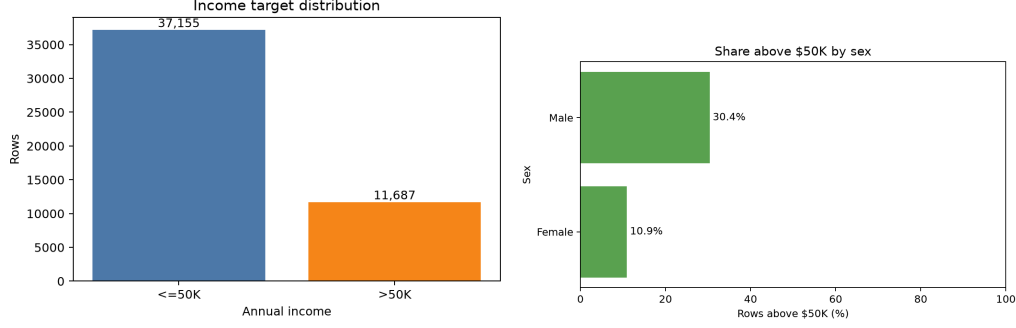


Figure 2: EDA of the cleaned Adult data. Left: target distribution. Right: observed proportion above \$50,000 by sex. These are descriptive dataset summaries, not predictions from a trained model.

Table 1: Experimental data and preprocessing setup.

Item	Specification
Dataset	UCI Adult Income
Training/test rows	32,561 / 16,281
Predictors	14 (6 numeric, 8 categorical)
Target	Income above \$50,000 (binary)
Sensitive attribute	sex, retained for evaluation
Missing categorical values	6,465 mapped to Missing
Encoder	Training-fitted one-hot encoding
Classifier	XGBoost, 300 trees, depth 6, learning rate 0.1
Decision rule	Positive if predicted probability ≥ 0.5
Random seed	42

ETL The ETL implementation reads `adult.data` and `adult.test` from `data/adult`. It removes surrounding whitespace, removes the period appended to labels in the test file, validates the target labels and numeric fields, and converts the target to 0/1. The documented ? values in categorical columns are mapped to an explicit `Missing` category rather than being silently dropped. This affected 6,465 values: 2,799 in `workclass`, 2,809 in `occupation`, and 857 in `native-country`. The cleaned training and test splits, combined dataset, and ETL metadata are written to `data/processed`.

EDA The exploratory analysis is descriptive and is performed on the cleaned combined dataset before model fitting. It reports descriptive statistics and creates visualizations for the target distribution, age by income class, income rates by education, numeric feature distributions, and documented missing-value counts. We also compare the target rate across the two observed `sex` categories as an initial descriptive fairness diagnostic; this comparison is not a model fairness result. In the cleaned data, 11,687 of 48,842 records are above \$50,000. The corresponding observed rates are 30.38% for males (9,918/32,650) and 10.93% for females (1,769/16,192).

3.3 Evaluation metrics

We evaluate each model on the held-out test split using both predictive performance and group fairness metrics. Predictive performance is summarized by accuracy, balanced accuracy, precision, recall, and F1 score. Accuracy is the fraction of correct predictions; balanced accuracy is the mean recall across the two income classes; precision is the fraction of predicted above-\$50,000 cases that are correct; recall is the fraction of actual above-\$50,000 cases that are identified; and F1 is the harmonic mean of precision and recall. Reporting both accuracy and balanced accuracy is useful because the target is not evenly distributed.

Fairness metrics are computed across the observed binary `sex` categories, using the same fixed prediction rule for every model. The demographic-parity difference is defined as

$$\max_{g \in \mathcal{G}} P(\hat{Y} = 1 \mid A = g) - \min_{g \in \mathcal{G}} P(\hat{Y} = 1 \mid A = g),$$

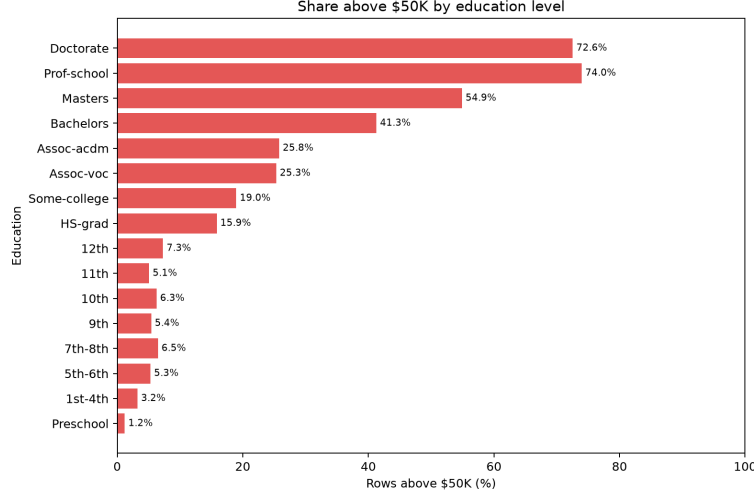


Figure 3: Observed proportion above \$50,000 by education category in the cleaned dataset.

Table 2: Descriptive statistics from the cleaned Adult dataset.

Group	Rows	Above \$50K	Rate (%)
Female	16,192	1,769	10.93
Male	32,650	9,918	30.38
All records	48,842	11,687	23.93

where A is sex and $\mathcal{G}$ is the set of observed group labels. It measures disparity in positive prediction rates, without conditioning on the true income label. The equalized-odds difference is reported as the larger of the across-group differences in true-positive rates and false-positive rates:

$$\max \left\{ \max_g \text{TPR}_g - \min_g \text{TPR}_g, \max_g \text{FPR}_g - \min_g \text{FPR}_g \right\}.$$

Thus, it measures disparity in both types of conditional classification outcomes. Lower values indicate smaller observed group differences for both fairness measures, while zero does not establish that a system is fair in all relevant senses.

The comparison table will include the baseline XGBoost model with sex, a model with sex removed, models with suspected proxy features ablated, and Fairlearn ExponentiatedGradient models constrained by demographic parity or equalized odds. The same predictive and fairness metrics will be reported for each variant so that feature removal, proxy removal, and algorithmic mitigation can be compared without collapsing their different objectives into a single score. We do not perform robustness or threshold analyses; the reported values correspond to the prespecified evaluation procedure and fixed prediction rule.

4 Results

4.1 Data preparation and descriptive audit

The ETL completed without dropping rows: all 48,842 records were retained, including the documented missing categorical values. The cleaned data contain 37,155 records at or below \$50,000 and 11,687 records above \$50,000. The descriptive target rate differs across the observed binary sex categories: 30.38% among males (9,918/32,650) and 10.93% among females (1,769/16,192). This is a property of the dataset and is not a prediction or a model fairness metric. It establishes the group labels and target distribution used in the subsequent held-out evaluation.

Table 3: Baseline XGBoost performance and fairness on the held-out test split ($n = 16,281$).

Metric	Value
Accuracy	0.8746
Balanced accuracy	0.7990
Precision	0.7786
Recall	0.6557
F1	0.7119
Demographic-parity difference	0.1710
Equalized-odds difference	0.0604

Table 4: Baseline group-level rates on the held-out test split.

Sex	Rows	Selection rate	TPR	FPR
Female	5,421	0.0849	0.6102	0.0207
Male	10,860	0.2559	0.6640	0.0811

4.2 Baseline performance and fairness

Table 3 reports the first held-out baseline result. On the 16,281-record test split, the model achieved 87.46% accuracy and 79.90% balanced accuracy. Precision was 77.86%, recall was 65.57%, and F1 was 71.19%. These values describe the unconstrained model that includes `sex`; they are not evidence that the model is fair or unfair without reference to the explicitly reported group metrics.

The baseline positive prediction rate was 8.49% for females and 25.59% for males, producing a demographic-parity difference of 0.1710. The true-positive rates were 0.6102 and 0.6640, respectively, while the false-positive rates were 0.0207 and 0.0811. The equalized-odds difference was therefore 0.0604, driven by the false-positive-rate gap rather than the true-positive-rate gap. These results are descriptive of this fixed split, model, and threshold; they do not establish causal discrimination or generalize beyond the dataset and audit design.

5 Discussion

6 Conclusion

References

- Kjersti Aas, Martin Jullum, and Anders Løland. Explaining individual predictions when features are dependent: More accurate approximations to shapley values. *Artificial Intelligence*, 298:103502, 2021. doi: 10.1016/j.artint.2021.103502. URL <https://arxiv.org/abs/1903.10464>.
- Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna Wallach. A reductions approach to fair classification. *arXiv preprint arXiv:1803.02453*, 2018. URL <https://arxiv.org/abs/1803.02453>.
- Solon Barocas, Moritz Hardt, and Arvind Narayanan. *Fairness and Machine Learning*. 2019. URL <https://fairmlbook.org/>.
- Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017. doi: 10.1089/big.2016.0047. URL <https://arxiv.org/abs/1703.00056>.
- Sam Corbett-Davies and Sharad Goel. The measure and mismeasure of fairness: A critical review of fairness criteria. In *Proceedings of the 2017 ACM Conference on Fairness, Accountability, and Transparency*, pages 5–15, 2017. doi: 10.1145/3178876.3186138.
- Anupam Datta, Matt Fredrikson, Gihyuk Ko, Piotr Mardziel, and Shayak Sen. Proxy non-discrimination in data-driven systems. *arXiv preprint arXiv:1707.08120*, 2017. URL <https://arxiv.org/abs/1707.08120>.
- Dheeru Dua and Casey Graff. Uci machine learning repository: Adult dataset, 2019. URL <https://archive.ics.uci.edu/dataset/2/adult>.

- Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. Fairness through awareness. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*, pages 214–226, 2012. doi: 10.1145/2090236.2090255. URL <https://arxiv.org/abs/1104.3913>.
- Moritz Hardt, Eric Price, and Nathan Srebro. Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems*, 2016. URL <https://arxiv.org/abs/1610.02413>.
- Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. Inherent trade-offs in the fair determination of risk scores. In *Proceedings of the 8th Innovations in Theoretical Computer Science Conference*, 2017. doi: 10.4230/LIPIcs.ITCS.2017.43. URL <https://arxiv.org/abs/1609.05807>.
- Matt J. Kusner, Joshua Loftus, Chris Russell, and Ricardo Silva. Counterfactual fairness. In *Advances in Neural Information Processing Systems*, 2017. URL <https://arxiv.org/abs/1703.06856>.
- Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *arXiv preprint arXiv:1705.07874*, 2017. URL <https://arxiv.org/abs/1705.07874>.
- David Madras, Elliot Creager, Toniann Pitassi, and Richard Zemel. Learning adversarially fair and transferable representations. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 3384–3393, 2018. URL <https://proceedings.mlr.press/v80/madras18a.html>.
- Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6):1–35, 2021. doi: 10.1145/3457607.
- Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P. Gummadi. Fairness constraints: Mechanisms for fair classification. In *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, volume 54 of *Proceedings of Machine Learning Research*, pages 962–970, 2017. URL <https://proceedings.mlr.press/v54/zafar17a.html>.